

Mayank Agrawal

amayank61@gmail.com | (585) 202-4243 | Rochester, NY | www.linkedin.com/in/mayank-ag2205

SKILLS

Programming Languages: Python, Java, C#, SQL, DAX, ASP.NET Core, HTML, CSS, Angular 12

Dashboards: Power BI, Tableau

Databases: MySQL, MongoDB, PostgreSQL, Neo4J, Microsoft Excel, Relational Database Management, Non - Relational Database Management, Data Warehouse, SAS, Pentaho Kettle, Data Modeling

Tools and Technologies: Microsoft Azure, AWS (EC2, S3, Lambda), Hadoop, Git, Microsoft Office, qGIS, ETL Data pipelines, NumPy, Pandas, Apache Spark, Snowflake, Jupyter

WORK EXPERIENCE

Teaching Assistant | *Rochester Institute of Technology* | *Rochester, NY*

Aug 2023 - Dec 2024

- Assisted in the design and implementation of hands-on assignments and projects to enhance student understanding of Java Database Connectivity (JDBC) concepts.
- Graded assignments and midterm exams, ensuring consistent and fair evaluation for over 120 students.
- Conducted regular office hours, providing one-on-one and group support to address student queries and clarify course material.

GCCIS Academic Tutor | *Rochester Institute of Technology* | *Rochester, NY*

Sep 2022 - Dec 2024

- Tutored undergraduate students in Python and Java for the 'Software Development and Problem Solving' course, enhancing their understanding of programming concepts.
- Provided academic support to over 200 undergraduate students, focusing on problem-solving concepts like Data structures, Algorithms, and Object-Oriented Programming.

Programmer Analyst | *Cognizant Technology Solutions* | *Pune, India*

Aug 2021 - Aug 2022

- Designed and developed interactive data dashboards using **Tableau** and **SQL**, enabling data-driven decision-making by visualizing customer buying behavior.
- Utilized **Python** for data preprocessing and transformation tasks, improving data quality and uncovering actionable insights from large datasets.
- Built and managed **ETL pipelines using Azure Data Factory**, automating data workflows and ensuring reliable, scalable data integration across systems.

Software Engineering Intern | *Cognizant Technology Solutions* | *Pune, India*

Jan 2021 - Jul 2021

- Interned as software developer trainee to develop core development skills like ASP.NET framework and Angular 12.
- Engineered 2 full stack web applications using ASP.NET core framework.
- Designed a web application that tracked live stock market data and calculated real-time stock profile values, collaborating with a team of 4 interns to enhance data accuracy and decision-making.

EDUCATION

Rochester Institute of Technology | *Rochester, NY*

Aug 2022 – Dec 2024

Master of Science, Information Technology and Analytics (GPA: 3.7/4)

Rajiv Gandhi Proudlyogiki Vishwavidyalaya | *India*

Jun 2017 – July 2021

Bachelor of Technology, Computer Science (GPA: 7.84/10)

PROJECTS

Microsoft Azure Dynamic ETL Pipeline for Sales Data

- Built a **dynamic ETL pipeline** using **Azure Data Factory (ADF)** to ingest sales data from a **REST API containing 10+ files and over 2 0,000 records**. Stored raw and ingested data in **Azure Data Lake Storage Gen2** using a structured folder hierarchy (bronze/silver/gold layers).
- Utilized **Azure Databricks with PySpark** for **data cleansing, transformation, and enrichment**. Handled missing values, schema evolution, and optimized data partitioning for performance.
- Built a **data warehouse in Azure Synapse Analytics**. Created **external tables, views, and schemas** directly referencing transformed data in the data lake.
- Enabled **Power BI connectivity** with Synapse for business intelligence and visualization. Delivered insightful dashboards with KPIs, trends, and drill-down analytics from the unified sales dataset.

Predictive Analysis of NHL Team Performance Using Machine Learning

- Designed and implemented machine learning models, including Logistic Regression, Random Forest, and Gradient Boosting, to predict high- vs. low-performing NHL teams based on in-game metrics.
- Extracted and engineered features from game and play-by-play data to represent critical performance metrics such as shot accuracy, penalty frequency, blocked shots, and power-play success rate.
- Created and interpreted advanced visualizations using PowerBI, including correlation heatmaps, pair plots, and residual plots, to gain insights into feature relationships and model performance.

CERTIFICATIONS

Microsoft Certified: Power BI Data Analyst Associate, Microsoft Certified: Azure Data Engineer Associate